

WESTERN SYDNEY
UNIVERSITY



Module 3

Solving a System of Linear Equations

A System of Linear Equations

$$A_{11}x_1 + A_{12}x_2 + A_{13}x_3 = B_1$$

$$A_{21}x_1 + A_{22}x_2 + A_{23}x_3 = B_2$$

$$A_{31}x_1 + A_{32}x_2 + A_{33}x_3 = B_3$$

$$AX = B$$

$$\text{where } A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{32} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \text{ and } B = \begin{bmatrix} B_1 \\ B_2 \\ B_3 \end{bmatrix}$$

$$A^{-1}AX = A^{-1}B$$

$$IX = A^{-1}B$$

$$X = A^{-1}B$$

Multiply both side by A^{-1}

Note that $A^{-1}A = I$

Note that $IX = X$

Left division operator, \

Left division is one of MATLAB's two kinds of array division

- Used to solve the matrix equation $AX=B$
 - A is a square matrix, X , B are column vectors
 - Solution is $X = A^{-1}B$

In MATLAB, solve by using left division operator ($\backslash$),
i.e.,

```
>> X = A \ B
```

When solving set of linear equations, use left division, not inverse, i.e., use $X=A \setminus B$ not $X=\text{inv}(A) * B$

Left division is:

- 2-3 times faster
- Often produces smaller error than `inv()`
- Sometimes `inv()` can produce erroneous results

Right division operator /

Right division is the other kind of MATLAB's array division

- Used to solve the matrix equation $XC=D$
 - C is a square matrix, X , D are row vectors
 - Solution is $X = DC^{-1}$

In MATLAB, solve by using right division operator ($/$), i.e.,

```
>> X = D / C
```